

Overview of the Æther-flow Framework

Exact-GR benchmark and source proposal

Abstract

The Æther-flow program has a physics objective and an artificial-intelligence research objective. The physics task is to determine whether a specified development of its source ontology can recover a declared part of general relativity without assuming that result, or whether an obstruction can be established within an explicit scope. The research-system task is to evaluate independently checkable progress on those questions. The current manuscripts provide a general-relativistic comparison model and a limited source proposal: a smooth four-dimensional arena with an untyped flow placeholder. This overview maps their contributions and sets out a bounded feasibility sequence. It distinguishes fixed-assumption audits from open construction, and successful derivations from counterexamples, scoped impossibility results, and unresolved searches. No source recovery theorem, universal no-go theorem, or completed system evaluation is reported.

1 Two questions, two levels of description

The Æther / Æther-flow framework asks whether observed spatial and temporal organization can be understood through a source structure not identified in advance with physical spacetime. The framework also makes a separate effective choice: use general relativity (GR) for the quantitative gravitational description. Keeping these two statements distinct is the organizing principle of the presentation.

The physical intent is an account of GR in which intrinsic ordered motion of a four-dimensional substrate produces the spacetime relations accessible to observers within it. The source is not intended as a wind through pre-existing three-dimensional space. Local spatial experience and S-time express the proposed connection between source organization and measurement. Preserving GR is the target of this construction, not evidence that the construction has already been achieved. The mathematical questions below test that hypothesis and admit a justified result against a specified candidate.

At source level, $\mathcal{A}$ denotes a smooth four-dimensional manifold and Φ_{src} is an untyped placeholder for a possible order or evolution structure. These objects do not yet supply a source metric, causal order, clock, matter model, or reconstruction map. At effective level, (M, g) is a Lorentzian spacetime with ordinary matter laws and operational definitions of measured quantities. There is no identification $\mathcal{A} = M$ or $\Phi_{\text{src}} = u$ between the source placeholder and an observer four-velocity.

The term *exact closure* means that the effective gravitational law is fixed to the corresponding GR law, rather than approximately matched in one limit. It does not mean that the source ontology is complete, that all matter has been specified, or that the physical origin of the Einstein equation has been explained.

2 Common effective statement

We use signature $(- + ++)$, $x^0 = ct$ in length-valued covariant coordinates, a physical matter action S_m , and a constant Λ . The benchmark action is

$$S_{\text{eff}}[g, \psi] = \frac{c^3}{16\pi G} \int_M d^4x \sqrt{-g} (R - 2\Lambda) + S_m[g, \psi], \quad (1)$$

with $G > 0$ and specified ordinary matter fields ψ . It yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad T_{\mu\nu} = -\frac{2c}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}. \quad (2)$$

The $\Lambda = 0$ specialization is used for isolated flat-background calculations. No independent source-flow field is varied in (1).

Standard vacuum light in geometric optics and ideal timelike clocks obey

$$g_{\mu\nu} k^\mu k^\nu = 0, \quad (3)$$

$$d\tau^2 = -\frac{1}{c^2} g_{\mu\nu} dx^\mu dx^\nu. \quad (4)$$

These are ordinary GR relations (Carroll, 1997). Observational identity further requires matching matter, constants, admissible solutions, initial and boundary data, and measurement procedures. An added source restriction on solutions would have to be included in the comparison; it cannot be hidden behind equality of the displayed field equations.

A constant Λ is equivalent to the specific vacuum tensor $T_{\mu\nu}^{(\Lambda)} = -\Lambda c^4 g_{\mu\nu} / (8\pi G)$, counted on one side of the equation only. A general conserved dark-energy component is a different matter model and needs its own specification. Cosmological acceleration is an allowed consequence of suitable GR models, not a source-to-GR derivation.

3 What each article contributes

The framework is developed across related expositions of one proposal. Their reading order is as follows.

Exact-Closure Note. The compact statement of the effective model and its conditional relationship to GR. It is the shortest route to the central distinction between adopted dynamics and a proposed source explanation.

Foundations. The first technical paper defines the narrow source assumptions. It separates the smooth source arena from spacetime and treats experienced spatial slices and S-time as reconstruction aims rather than derived structures.

Dynamics. The second technical paper fixes the action and stress–energy units, derives the effective equation, and specifies matter coupling, conservation, trajectories, and the weak-field limit. It distinguishes an arbitrary matter template from a fully specified physical model.

Consistency. The third technical paper gives the standard gauge and constraint analysis of the adopted gravitational sector. Its explicit stability calculation concerns weak radiative perturbations about Minkowski vacuum, not every GR background or a source theory that has not yet been defined.

Relativistic Recovery. The fourth technical paper derives representative observable relations: Newtonian acceleration, null propagation, static redshift, proper time, local special relativity, and homogeneous cosmology. Its results are inherited from the corresponding GR models under stated conditions.

Geometry. The fifth technical paper develops observer rest spaces and congruence kinematics. It derives Raychaudhuri evolution with consistent units, distinguishes static support from free fall, gives a zero-vorticity frame-dragging example, and separates shear, curvature, and wave strain.

Synthesis article. The synthesis article gives a self-contained conceptual account of the source proposal and the effective interpretation. It explains the explanatory gap and relates the approach to other gravitational research without claiming a new low-energy field law.

A reader seeking the main conceptual argument can begin with the synthesis, which also specifies the proposed AI evaluation protocol, and use the technical papers for the corresponding calculations. The canonical comprehensive manuscript contains these eight expositions in one document; the standalone versions retain the same scientific content for focused reading.

4 Results and shared limits

The effective action has the usual local GR gravitational degrees of freedom. The Arnowitt–Deser–Misner (ADM) count and the Minkowski transverse-traceless analysis yield two radiative gravitational polarizations (Arnowitt et al., 1962). No additional *Æther-flow* vector or scalar gravitational mode has been introduced. This does not count matter modes or prove stability on arbitrary backgrounds.

The geometry paper uses standard observer kinematics: expansion, shear, vorticity, and acceleration (Ellis & van Elst, 1999). These are properties of a metric and chosen observer family. They neither define a source flow nor provide a unique taxonomy of gravity. In particular, congruence vorticity is not synonymous with frame dragging, and zero shear does not imply absence of tidal curvature.

The shared source limit is equally precise. The papers do not derive Lorentzian signature, metric dynamics, ordinary matter, operational clocks, or measured constants from $\mathcal{A}$ and Φ_{src} . They also do not establish that a literal substrate exists. The absence of such results is not a proof that all possible source completions fail; it identifies which argument is still required.

5 Relation to other approaches

The effective benchmark belongs to standard GR, not to a preferred-structure modification. Einstein-aether theory adds a dynamical timelike vector, whereas Hořava’s proposal changes the short-distance spacetime scaling structure (Hořava, 2009; Jacobson & Mattingly, 2001). Sharing words such as aether or flow does not make these actions equivalent to (1).

The source ambition is closer in subject to approaches that seek an explanation for relativistic dynamics from other assumptions. Examples include thermodynamic reasoning, induced gravity, and consistency-based self-coupling (Deser, 1970; Jacobson, 1995; Visser, 2002). Each has its own premises. Their existence does not supply the missing premises of this ontology, and a successful result in one approach cannot be transferred here by analogy. The detailed comparison is reserved for the synthesis rather than repeated as a large citation cluster in every article.

6 Use of the benchmark

The effective GR model can remain a comparison target while source work proceeds. If a proposed extension fails, one may set that extension aside and retain the independent benchmark. This is a research decision, not a proof that source assumptions are immune to failure. Any assumption implicated by a failed calculation

must be reconsidered on its own merits.

7 A bounded feasibility sequence

The intended research infrastructure is Sys4AI, an agentic AI framework for coordinating roles, tasks, and verification. The manuscripts specify the research question and comparison physics; they do not report a running evaluation implementation or measured agent performance. The two-track intent is to investigate the source proposal and to assess the reliability of the research process used to investigate it.

The initial feasibility task is narrower than deriving all of GR at once. First, define a versioned source model and one reconstruction target, such as local causal geometry or a clock relation. State every additional assumption and exhibit an admissible source example or the relevant existence argument. Then seek a derivation, a counterexample to the stated implication, or an obstruction within a declared class. Preserve an unresolved outcome when none is established.

The next stage depends on the result. A verified geometric construction may support a separate dynamics or matter-coupling task. A negative result should identify the premise or reconstruction restriction responsible and delimit which alternatives remain open. An unfinished argument should identify its missing step. No stage receives credit for importing the desired target into its assumptions.

Each completed task should yield a self-contained claim, an assumption record, the argument or witness, supporting calculations where needed, and an independent checking record. Resource use and human intervention belong in the AI evaluation record. A valid result is assessed at its actual scope; a partial geometric construction does not become a full GR derivation because later stages are planned.

Two modes organize this sequence. A fixed-assumption audit evaluates the proposal as given. Open construction permits explicit additions and assesses the strengthened proposal. The synthesis article provides the fuller evaluation protocol, including controls, comparable budgets, and the distinction between a base model and the entire research system. This is a proposed procedure, not evidence that these stages have already been completed.

8 Conclusion

The framework presents a definite effective gravitational model and a deliberately limited source proposal. Its technical value lies in explicit conventions, calculations, observer choices, and limits. Its open question is whether a specified source theory can reproduce those structures without assuming them in disguised form. The bounded next step is a versioned source specification and a checkable claim. The physics and AI tracks can both learn from a valid obstruction, while an unfinished search remains unfinished. Performance claims await an implemented protocol, recorded runs, and independent assessment.

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